

1. Ramesh's basic salary is input through the keyboard. His dearness allowance is 40% of basic salary, and house rent allowance is 20% of basic salary. Write a program to calculate his gross salary.

```
In [5]: salary = eval(input("Enter a basic Salary :"))
dearness = salary*40/100
house_rent = salary*20/100
gross_salary = salary + house_rent + dearness
print("Gross Salary:", gross_salary)
```

Gross Salary: 64000.0

2. The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches and centimeters.

```
In [8]: distance = eval(input("Enter a number two distance Cites:"))
meter = distance * 1000
centimeter = distance * 10000
inch = distance * 39370.1
feet = distance * 3280.84
print(f"Distance in Meter {meter} , Centimeter {centimeter} , Inch {inch} , Feet
```

Distance in Meter 5000 , Centimeter 50000 , Inch 196850.5 , Feet 16404.2

3. If the marks obtained by a student in five different subjects are input through the keyboard, find out the aggregate marks and percentage marks obtained by the student. Assume that the maximum marks that can be obtained by a student in each subject is 100.

```
In [4]: math = eval(input('Mathematics : '))
science = eval(input('Science : '))
english = eval(input('English: '))
hindi = eval(input('Hindi: '))
computer = eval(input('Computer: '))

total_per = (math+science+english+hindi+computer)/500 *100

print("Total Percentage of Student :", total_per)
```

Total Percentage of Student : 88.4

4. Temperature of a city in Fahrenheit degrees is input through the keyboard. Write a program to convert this temperature into Centigrade degrees.

```
In [3]: fahrenheit = eval(input("Enter a Fahrenheit :"))
centigrade = (fahrenheit - 32)*5/9

print("Convert Fahrenheit degrees to Centigrade degrees ", round(centigrade, 2))
```

Convert Fahrenheit degrees to Centigrade degrees 1.11

5. The length & breadth of a rectangle and radius of a circle are input through the keyboard. Write a program to calculate the area & perimeter of the rectangle, and the area & circumference of the circle.

```
In [5]: l = eval(input("Enter a Length :"))
b = eval(input("Enter a Bright :"))
r = eval(input("Enter a radius :"))

area_of_ractangle = l * b
perameter = 2*(l+b)
area_of_circle = 3.14*r*r
circumference = 2 *3.14*r

print("Area Of ractngle :",area_of_ractangle)
print("Area of Peramerter :",perameter )
print("Area of Circle:",area_of_circle)
print("Area of Circumfarence:",circumference )
```

```
Area Of ractngle : 1035
Area of Peramerter : 136
Area of Circle 9847.04
Area of Circumfarence 351.68
```

6. Two numbers are input through the keyboard into two locations C and D. Write a program to interchange the contents of C and D.

```
In [ ]: a = 3
b = 4
```

7. If a five-digit number is input through the keyboard, write a program to reverse the number.

```
In [8]: n = 12345
reverse = 0

while(n>0):
    digit = n%10
    reverse = reverse*10+digit
    n= n//10
print("Revese the number :",reverse)
```

```
Revese the number : 54321
```

6. Two numbers are input through the keyboard into two locations C and D. Write a program to interchange the contents of C and D.

```
In [3]: c = input("Enter a C Location : ")
d = input("Enter a D Location : ")
c,d = d,c

print(c)
print (d)
```

45
34

9. In a town, the percentage of men is 52. The percentage of total literacy is 48. If total percentage of literate men is 35 of the total population, write a program to find the total number of illiterate men and women if the population of the town is 80,000.

```
In [10]: man = 80000 * 52 / 100
litr_all = 80000 * 48 / 100
man_lit = 80000 * 35 / 100
women_lit = litr_all - man_lit
women = 80000 * 48 / 100
print("ans man illiterate ", man - man_lit)
print("ans woman illiterate ", women - women_lit)
```

```
ans man illiterate 13600.0
ans woman illiterate 28000.0
```

8. If a four-digit number is input through the keyboard, write a program to obtain the sum of the first and last digit of this number.

```
In [3]: n = eval(input("Enter a number"))

first = n // 1000
last = n % 10
s = first + last
print(s)
```

7

10. A cashier has currency notes of denominations 10, 50 and 100. If the amount to be withdrawn is input through the keyboard in hundreds, find the total number of currency notes of each denomination the cashier will have to give to the withdrawer.

```
In [3]: amount = eval(input("Enter a amount to withdraw :"))

notes100 = amount // 100
amount = amount % 100

notes50 = amount // 50
amount = amount % 50

notes10 = amount // 10

print("100 rupee notes =", notes100)
print("50 rupee notes =", notes50)
print("10 rupee notes =", notes10)
```

```
100 rupee notes = 30
50 rupee notes = 1
10 rupee notes = 2
```

In []: